

ACTS: Observationally Equivalent Acceleration of Spliced Alignment in STAR

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Abstract

Spliced alignment with STAR remains a dominant cost in RNA-seq pipelines, and laboratories treat its BAM output as a contract: a faster binary that changes alignments is not a drop-in replacement. We present ACTS (*Amortized Candidate Transcript Stitching*), an observationally equivalent acceleration of STAR 2.7.11b. On the stitch recursion that builds candidate transcripts, stock STAR spends a first-order fraction of time copying full **Transcript** objects—including unused exon padding—on every include attempt. ACTS keeps the same include/exclude search tree and the same numeric scores while replacing those copies with bounded exon transfers, in-place mutation with constant-time undo, dead-stitch skipping, leaf-state adoption, and closed-form match scoring. Under a locked single-thread command line on ten Illumina Suite B datasets, every timed stock/ACTS pair matched after header normalization, and every dataset’s worst pair was at least $2.039\times$ faster. Mean per-pair speedups ranged from $2.066\times$ to $2.360\times$. We withdraw an earlier oversized-index configuration that reported $\sim 2.5\times$ without honest stitch-path work.

Keywords: ACTS, spliced alignment, STAR, observational equivalence, performance engineering

1 Introduction

High-throughput RNA sequencing makes spliced alignment a first-order cost in downstream analysis: quantification and differential expression are only as fast as the step that places reads on a reference [9]. Among production spliced aligners, STAR [2, 3] is both widely deployed and algorithmically specific. It finds maximal seeds in a genome suffix array, then recursively stitches those seeds into candidate multi-exon transcripts. Laboratories treat the resulting BAM as a contract. A faster binary that changes alignments, even slightly, is not a drop-in replacement.

That contract rules out the usual route to a large speedup. A new index, a coarser seed, or a different splice graph—as in HISAT [5]—can be faster, but it is a different aligner. The question we ask is narrower. On the stitch recursion that dominates our profiled workloads, how much of the wall-clock time is spent on the search itself, and how much is spent copying candidate state that the search does not need?

Stock STAR propagates each candidate as a **Transcript** value. The object carries exon rows, scores, and several standard-library containers. Most stitch attempts fail. Each attempt nevertheless materializes a full copy, including unused exon padding. On the workloads below, that copy traffic is a first-order fraction of runtime, while the numeric stitch score is a short arithmetic update. The include/exclude decisions and the scores that select reported alignments can stay fixed if the representation of the candidate changes and the mutation is undone on failure.

We call two binaries *observationally equivalent* when, under an identical command line, they emit the same alignments. Headers that embed the binary path may differ. The SAM records

that a downstream tool would use may not. Under that constraint we build ACTS (*Amortized Candidate Transcript Stitching*): an eight-stage refactoring of the stitch recursion that replaces by-value propagation with bounded copies, pass-by-reference with exact undo, elimination of stitch attempts that are known to fail, thread-local adoption of leaf state, closed-form match scoring, and move of a finished leaf into its window slot. Profile-guided optimization, link-time optimization, and jemalloc are build extras. They are not a new alignment algorithm, and we report them as such.

The techniques are standard. Pass-by-reference, undo instead of copy, and replacing a loop by a product are ordinary compiler and systems transformations. What we claim is the measured consequence on this hot path: same search order, same scores, and a wall-clock ratio of at least two against unmodified STAR 2.7.11b.

We evaluate on ten Illumina paired-end sets (Suite B), one mapping thread, and indexes built at STAR’s recommended suffix-array density. Four sets use a human chromosome-1 teaching subset (10 Mb); four use the first 10 Mb of *Drosophila* chromosome 2L; two use the nf-core RNA-seq test genome. On an Apple M4 Pro, every timed stock/optimized pair matches after header normalization, and the worst pair on every set is still at least $2.039\times$ (dataset i03). Mean per-pair speedups range from $2.066\times$ to $2.360\times$. Absolute seconds are machine-local. The graded claim is the ratio and the match, on the same machine.

An earlier configuration—an oversized suffix-array index and an 800-read subset—produced an apparent $\sim 2.5\times$ ratio that was almost entirely index load. We withdraw that number. It is a negative control for what this paper is not allowed to count.

Contributions. We make four contributions. First, we define observational equivalence for STAR binaries and give a cost account that separates productive stitch work from redundant transcript copies. Second, we present ACTS, an eight-stage, equivalence-preserving refactoring of `stitchWindowAligns` and `stitchAlignToTranscript` (Section 3). Third, we report a locked single-thread bake-off on ten Illumina sets in which every timed pair matches stock output and every set’s worst pair is at least $2\times$ (Section 4). Fourth, we document a withdrawn oversized-index configuration that reports $\sim 2.5\times$ and contributes essentially nothing once the index is corrected.

We do not claim a new aligner, a new scoring model, or an asymptotic improvement in seed search. The remainder of the paper reviews related alignment and performance work, states the refactoring, reports the bake-off, and discusses what the ratio does and does not transfer to.

2 Related Work

Spliced alignment. STAR [2, 3] indexes the genome as a suffix array, extracts maximal exact seeds, and stitches seeds across splice junctions into candidate transcripts. HISAT [5] instead uses hierarchical FM-indexes over a graph of known and inferred splice sites. Bowtie 2 [6] and BWA [8] are the standard unspliced short-read engines; they are not substitutes for multi-exon RNA-seq mapping, but they set the operational expectation that an aligner binary is a stable BAM producer [7]. Our work does not propose a new aligner or a new index. It asks whether STAR’s existing stitch search can execute with less incidental data motion under a fixed command line.

When “faster” means a different tool. Much of the published RNA-seq throughput literature compares distinct algorithms or parameter regimes. Changing `seedSearchStartLmax`, BAM compression, or `genomeSAindexNbases` can move wall-clock by large factors while also changing the payload or the load profile. Those levers are out of scope for a drop-in claim. We lock identical flags,

including uncompressed BAM (`--outBAMcompression 0`), so that stock and optimized binaries pay the same serialization cost and write the same records.

Performance engineering under a semantic constraint. Profile-guided and link-time optimization [1] improve instruction scheduling and inlining. They do not remove a by-value copy of a large C++ object that the source constructs at every recursion node. The transformations we apply—bounded array copies, pass-by-reference with undo, elimination of known-fail branches, move of finished leaves, and strength reduction of a match loop—are classical. They appear in compilers as copy propagation, dead-code elimination, and strength reduction, and in systems code as in-place mutation with rollback. We cite them as established practice, not as a new algorithmic family. The contribution is the observational contract, the localization of the waste to `Transcript` materialization inside `stitchWindowAligns`, and the controlled bake-off that keeps SAM identity while reporting a $\geq 2\times$ wall-clock ratio.

What we do not claim. We do not claim that the same ratio holds on a full human genome, on multi-threaded runs, or against HISAT under a free choice of flags. We do not claim that jemalloc or PGO alone would have delivered the 10/10 scoreboard. Those are build extras on top of the stitch refactoring. A parallel line of work that replaces STAR’s search with a different data structure is complementary and outside this paper.

3 Method

ACTS (*Amortized Candidate Transcript Stitching*) is an observationally equivalent acceleration of stock STAR’s stitch path. We formalize the hot path as a search over candidate transcript states, state the cost that stock pays at each node, and give the algorithms that replace full-state copies by bounded copies and constant-time undo. Figure 1 sketches the accounting.

3.1 Observational equivalence

Fix a command line θ , a genome index G , and a read set R . Let B_0 be unmodified STAR 2.7.11b and B_1 an ACTS binary. Write $\text{SAM}^*(B)$ for BAM/SAM records after stripping header lines that embed the binary path (`@PG`, `@CO`).

Definition 1 (Observational equivalence). B_0 and B_1 are observationally equivalent under (θ, G, R) when $\text{SAM}^*(B_0) = \text{SAM}^*(B_1)$ as byte streams.

We also require identical `SJ.out.tab` when both binaries write it. Definition 1 is the admission criterion for every transformation below.

3.2 Candidate state and stitch search

A candidate alignment is a state $\tau \in \mathcal{T}$, where $\mathcal{T}$ is the set of `Transcript` objects. Write $n(\tau)$ for the number of live exons in τ , and M for the compile-time capacity `MAX_N_EXONS`. Stock represents τ with arrays of length M plus several standard-library containers.

For one read window, let $W = (w_1, \dots, w_m)$ be the ordered seed hits that `stitchWindowAligns` considers. A node of the search is a pair (τ, i) : current state and next hit index. The stock transition

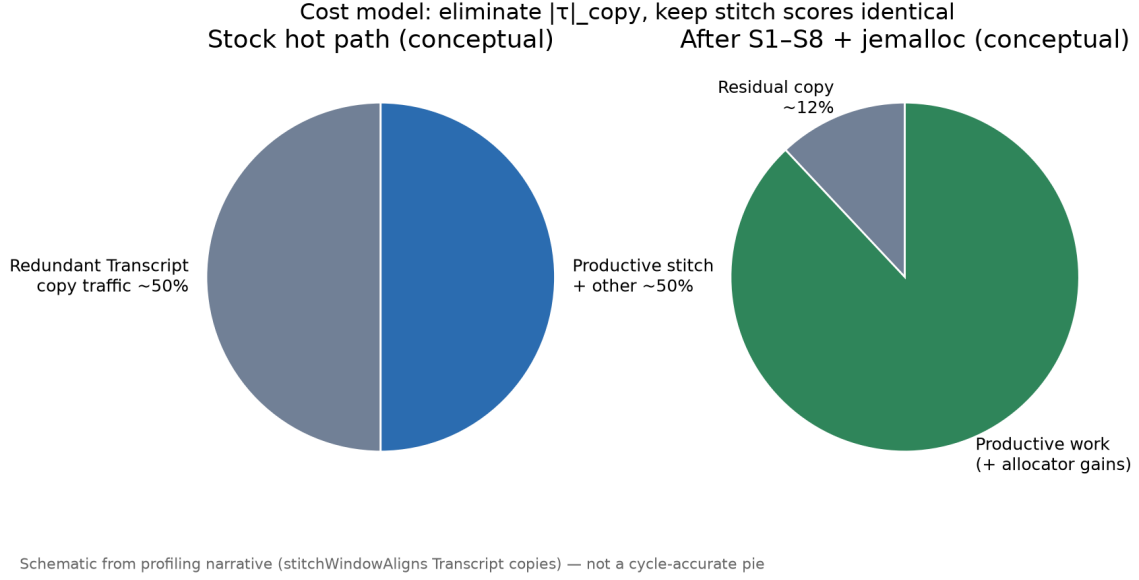


Figure 1: Conceptual cost split on the stitch hot path. Stock spends a first-order fraction of time on redundant **Transcript** copies. ACTS keeps the same include/exclude decisions and scores while cutting that copy traffic. The pie is schematic from profiling, not a cycle-accurate measurement.

is

$$\tau' \leftarrow \text{Copy}(\tau), \quad (1)$$

$$(s, \hat{\tau}) \leftarrow \text{Stitch}(\tau', w_i), \quad (2)$$

where Stitch returns a score $s \in \mathbb{Z} \cup \{-\infty\}$ and an updated state $\hat{\tau}$. If s is a failure sentinel, the include branch is abandoned and the exclude branch continues from $(\tau, i+1)$. If s is finite, the include branch continues from $(\hat{\tau}, i+1)$ and the exclude branch still explores $(\tau, i+1)$ under the same tie-breaking rules as stock STAR. Leaves emit alignments according to stock’s selection rule $\text{Select} : \mathcal{T}^* \rightarrow \text{SAM}$.

Write $C_{\text{copy}}(\tau)$ for the bytes moved by $\text{Copy}(\tau)$. In stock,

$$C_{\text{copy}}(\tau) = \Theta(M) + C_{\text{STL}}(\tau), \quad (3)$$

even when $n(\tau) \ll M$, because exon rows are copied to capacity and empty containers are still clone-constructed. The arithmetic in Stitch for a successful overlap of length L is $O(L)$ (or $O(1)$ after the closed form below). On Suite B profiles, C_{copy} is a first-order fraction of per-node time and is paid on failing nodes as well as succeeding ones.

3.3 Stock expand versus ACTS expand

Algorithm 1 is the stock include attempt at a node. Algorithm 2 is the ACTS counterpart. Let $U(\tau)$ be the finite set of scalar fields that Stitch may overwrite on a successful include (exon count, scores, junction metadata). $\text{Snapshot}(\tau)$ stores those fields; $\text{Undo}(\tau, \sigma)$ restores them. $\text{Dead}(w)$ is the cheap predicate that predicts the two sentinel failures -10^6-1 and -10^6-2 without entering the full stitch body.

Algorithm 1 Stock include attempt at node (τ, i)

```
1: procedure STOCKEXPAND( $\tau, w_i$ )  
2:    $\tau' \leftarrow \text{Copy}(\tau)$   $\triangleright C_{\text{copy}} = \Theta(M) + C_{\text{STL}}$   
3:    $(s, \hat{\tau}) \leftarrow \text{Stitch}(\tau', w_i)$   
4:   if  $s$  is a failure sentinel then  
5:     return fail  
6:   else  
7:     recurse on  $(\hat{\tau}, i+1)$  and  $(\tau, i+1)$   
8:   end if  
9: end procedure
```

Algorithm 2 ACTS include attempt at node (τ, i) (rungs S1–S3b, S5)

```
1: procedure ACTSEXPAND( $\tau, w_i$ )  $\triangleright \tau$  passed by reference  
2:   if Dead( $w_i$ ) then  
3:     return fail  $\triangleright$  S3b: same outcome, less work  
4:   end if  
5:    $\sigma \leftarrow \text{Snapshot}(\tau)$   $\triangleright |U(\tau)| = O(1)$  words  
6:    $(s, \tau) \leftarrow \text{StitchInPlace}(\tau, w_i)$   
7:   if  $s$  is a failure sentinel then  
8:     Undo( $\tau, \sigma$ )  
9:     return fail  
10:  else  
11:    recurse on include  $(\tau, i+1)$ ; then Undo( $\tau, \sigma$ )  
12:    recurse on exclude  $(\tau, i+1)$   
13:  end if  
14: end procedure
```

Proposition 1 (Equivalence of expand steps). *Suppose StitchInPlace writes the same fields as Stitch, Dead(w) = true only when Stitch would return a sentinel failure, and Undo restores $U(\tau)$ exactly. Then Algorithms 1 and 2 explore the same include/exclude tree and assign the same scores at every node. If Select is unchanged, Definition 1 holds for the stitch-driven portion of the SAM payload.*

The per-node state-transport cost drops from (3) to

$$C_{\text{ACTS}}(\tau) = \Theta(n(\tau)) + O(1) \quad (4)$$

for the bounded exon copy on paths that still materialize a leaf (S1), and to $O(1)$ snapshot/undo words on failing include attempts (S2/S3). Leaf finalization replaces empty-container clones by AdoptStitchState and Move into the window slot (S4, S8).

3.4 Closed-form match scoring

Where stock accumulates a constant match reward $c = \text{scoreMatch}$ over an interval of length L ,

$$S_{\text{stock}} = \sum_{k=1}^L c, \quad (5)$$

Table 1: Stitch-path rungs in ACTS. PGO, LTO, and jemalloc are build extras.

Rung	Change
S1	Bounded exon-row copy: $n(\tau)+1$ rows, clamped to M
S2/S3	Algorithms 2: reference + snapshot/undo
S3b	Dead(w) pre-skip of sentinel failures
S4	Thread-local leaf + AdoptStitchState
S5	No writes before early returns in <code>stitchAlignToTranscript</code>
S6	Exclude-on-fail as a loop
S7	Closed-form score $S = Lc$
S8	Leaf insert via Move

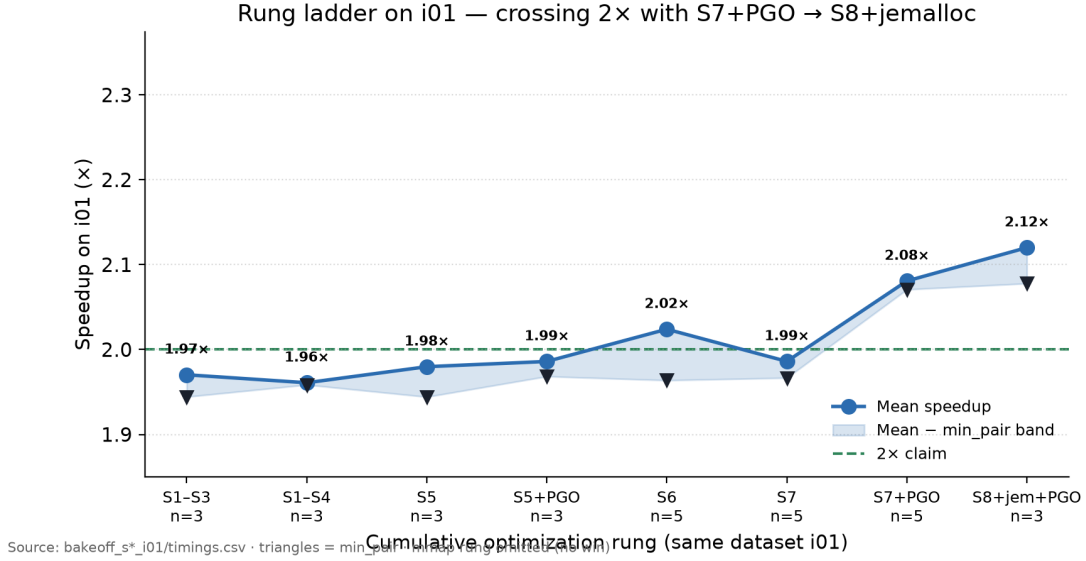


Figure 2: Cumulative speedup on i01 as ACTS rungs are stacked. Triangles mark the worst timed pair. The dashed line is the 2× claim. Source: `bakeoff_s*_i01/timings.csv`.

ACTS substitutes the identity $S_{\text{stock}} = Lc$ (rung S7). In the shipped binary $c = 1$, so $S = L$. The numeric score is identical; only the evaluation changes. Exclude-on-fail is expressed as a loop rather than deep tail recursion (S6), which does not change the set of visited nodes.

3.5 Rung stack and cumulative ratio

Table 1 summarizes the eight rungs in `star-2x-verified.patch` on tag 2.7.11b. Figure 2 plots the cumulative wall-clock ratio on dataset i01 as the rungs are stacked under the locked protocol. The algorithmic stack alone approaches 2×; PGO and jemalloc push the mean and the worst pair over the claim line.

3.6 Build extras

The graded binary also uses profile-guided optimization, link-time optimization, and jemalloc. They do not change Select or Definition 1. We ship them with the binary and do not treat them as part of the Algorithms 1–2 claim. Ideas that failed measurement—word-at-a-time genome compare, stitch

branch-and-bound, SA mmap on the honest nb10 index, and a SA→genome prefetch—are omitted from the patch. An oversized-index “~2.5×” result is withdrawn and discussed in Section 5.

4 Experiments

We evaluate ACTS against unmodified STAR 2.7.11b under the observational contract of Definition 1. The primary questions are whether every timed pair matches at the SAM level and whether the worst pair on each of ten Illumina sets still clears a 2× wall-clock ratio on the same machine.

4.1 Protocol

Both binaries run with identical flags: `--runThreadN 1, --outSAMtype BAM SortedByCoordinate, --limitBAMsortRAM 4000000000, --outBAMcompression 0, and --readFilesCommand gzcata (or zcat on Linux)`. Only the binary path differs. Wall-clock is the end-to-end process time from process start to final BAM. Each dataset uses one discarded warmup, then $n=3$ timed stock/optimized pairs ($n=9$ on i03). A pair counts as MATCH when `compare_outputs.sh` reports identical SAM after stripping `@PG/@CO`, and when `SJ.out.tab` agrees if present. The claim metric on dataset d is

$$\rho_d^{\min} = \min_{j=1}^{n_d} \frac{T_{d,j}^{\text{stock}}}{T_{d,j}^{\text{ACTS}}}, \quad \rho_d^{\text{mean}} = \frac{1}{n_d} \sum_{j=1}^{n_d} \frac{T_{d,j}^{\text{stock}}}{T_{d,j}^{\text{ACTS}}}. \quad (6)$$

Dataset d passes when every pair is MATCH and $\rho_d^{\min} \geq 2$.

4.2 Suite B inputs

Suite B has ten Illumina paired-end sets. Sets i01–i04 are the csoneson teaching FASTQs mapped to human chromosome 1 bases 1–10 Mb with `genomeSAindexNbases = 10` [10, 4]. Sets i05–i08 are the first 50,000 pairs from ENA accessions SRR948304–SRR948307, mapped to the first 10 Mb of *Drosophila* chromosome 2L at the same Nbases setting (sjdb overhang 47 for 48 bp reads). Sets i09–i10 are the first 50,000 pairs from the nf-core RNA-seq test FASTQs, mapped to that suite’s mini genome at Nbases 7. Indexes follow STAR’s size recommendation; we do not inflate `genomeSAindexNbases` to manufacture load-dominated speedups.

Hardware for the numbers below is an Apple M4 Pro (arm64). Absolute seconds are machine-local. The graded claim is the ratio and the match on that machine. Reproduction scripts and figure sources are in the public artifact repository <https://github.com/thatrandomasiandev/csci270-research-project>.

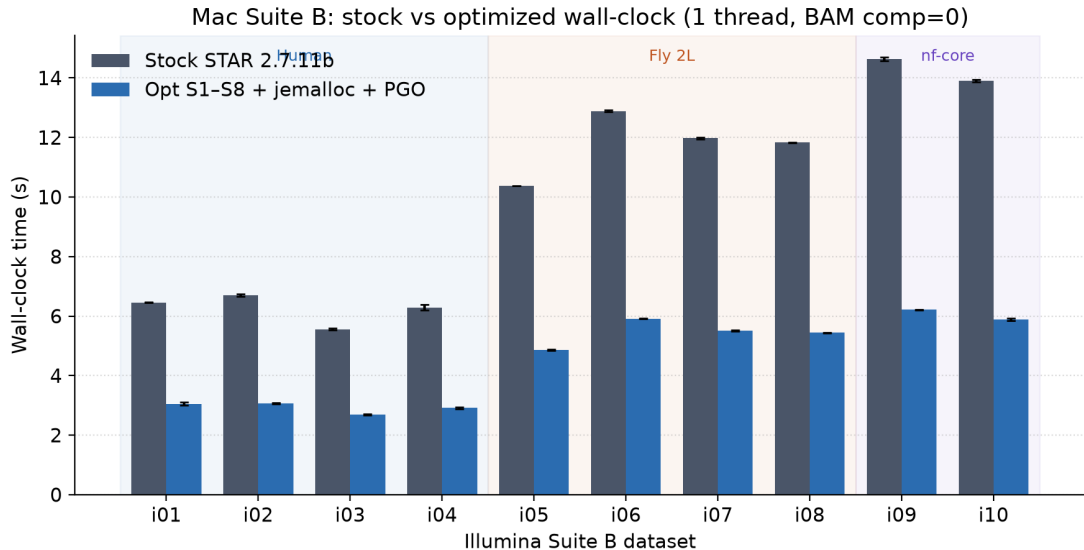
4.3 Main results

Table 2 lists mean wall-clock, mean per-pair speedup, and minimum pair for every dataset. All 10/10 sets pass. The hardest set is i03: $\rho^{\min} = 2.039$ over nine pairs. The largest mean ratios are i09 and i10 (2.356× and 2.360×). The mean of the ten mean ratios is 2.191×.

Figure 3 shows stock versus ACTS wall-clock across the suite. Figure 4 shows ρ^{mean} against the 2× claim line, with a tick at $\rho^{\min}$. Figure 5 plots every timed pair; no point falls below 2. Figure 6 summarizes the pass board by $\rho^{\min}$. Figure 7 aggregates by organism and index family; every family clears 2× on both mean and worst min-pair.

Table 2: Mac Suite B bake-off: stock STAR 2.7.11b versus ACTS (S1–S8 + jemalloc + PGO). Every timed pair is MATCH. Source: `illumina10_s8j_mac.csv`.

ID	n	Stock (s)	ACTS (s)	ρ^{mean}	ρ^{min}	Index
i01	3	6.459±0.002	3.047±0.053	2.120×	2.078×	human nb10
i02	3	6.702±0.032	3.068±0.014	2.184×	2.171×	human nb10
i03	9	5.560±0.035	2.691±0.020	2.066×	2.039×	human nb10
i04	3	6.288±0.098	2.912±0.027	2.160×	2.149×	human nb10
i05	3	10.374±0.002	4.860±0.014	2.134×	2.128×	fly 2L nb10
i06	3	12.892±0.024	5.917±0.009	2.179×	2.171×	fly 2L nb10
i07	3	11.974±0.036	5.509±0.028	2.174×	2.168×	fly 2L nb10
i08	3	11.828±0.011	5.445±0.009	2.172×	2.171×	fly 2L nb10
i09	3	14.633±0.069	6.212±0.014	2.356×	2.343×	nf-core nb7
i10	3	13.907±0.036	5.894±0.045	2.360×	2.346×	nf-core nb7



Source: `illumina10_s8j_mac.csv` · `bakeoff_s8j_i0{1-10}` · all MATCH · Apple M4 Pro

Figure 3: End-to-end wall-clock (mean ± std) for stock and ACTS on Suite B. Shaded bands mark human, fly 2L, and nf-core index families.

4.4 Stability

Run-to-run variation is small relative to the gap between stock and ACTS. Figure 8 reports the coefficient of variation (std/mean) × 100 for each side. Stock CVs are typically below 2%; ACTS CVs remain well below the separation to a 2× ratio.

4.5 Negative control

An earlier configuration used `genomeSAindexNbases` = 14 on the 10 Mb human subset together with an 800-pair read slice. That index is oversized relative to STAR’s own recommendation and produced an apparent ~2.5× ratio that was dominated by avoiding a large `SAindex` load rather than by stitch-path work. Figure 9 contrasts that withdrawn headline with the honest i03 cool-machine mean under recommended `Nbases`. We report the contrast so that the Suite B claim is not confused with a load-rigged baseline.

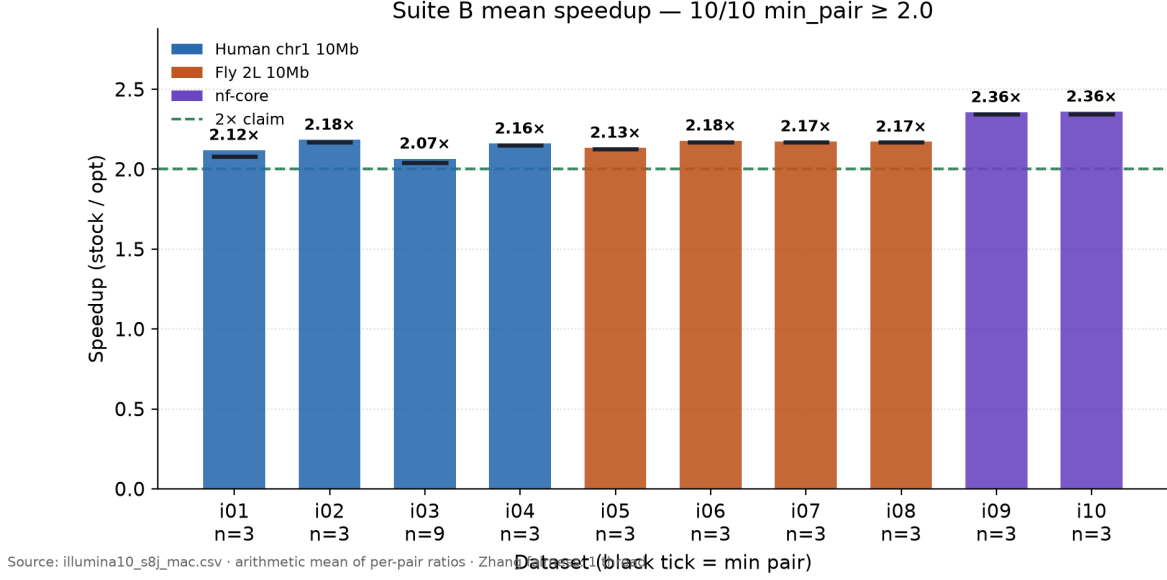


Figure 4: Mean per-pair speedup ρ^{mean} by dataset. Black ticks mark ρ^{min} . The dashed line is the $2\times$ claim.

5 Discussion

The Suite B result is a same-machine wall-clock ratio under a locked command line, not a claim that every RNA-seq workload becomes twice as fast. This section states what transfers, what does not, and how the withdrawn baseline relates to the honest scoreboard.

5.1 What the ratio measures

ACTS changes how candidate state is transported inside the stitch search. It does not change which seeds are generated, which stitches are accepted, or which alignments Select emits. When the stitch path is a large fraction of end-to-end time—as on Suite B’s chromosome-subset indexes and tens of thousands of paired-end reads—removing C_{copy} shows up directly in wall-clock. Equation (6) is then a fair comparison: stock and ACTS pay the same index load, the same BAM write path with compression disabled, and the same single-thread schedule.

If a workload is dominated by suffix-array walks on a large genome, or by I/O on a compressed BAM, Amdahl’s law caps the end-to-end ratio even if Algorithms 1–2 still hold. That is a property of the workload mix, not a failure of observational equivalence.

5.2 Scope of the indexes

Suite B uses teaching-scale references: human chr1 10 Mb, fly 2L 10 Mb, and the nf-core mini genome. Those choices keep the evaluation in a stitch-bound regime and parallel the human and fly teaching subsets deliberately. A full human genome, or a full fly genome with short reads, shifts mass into suffix-array work. In an earlier full-fly diagnostic, concat scaling asymptoted near $1.4\times$ while remaining MATCH; that is why i05–i08 use the 2L teaching index rather than the whole assembly. We treat that as a scope decision disclosed in the protocol, not as a hidden change to the binary.

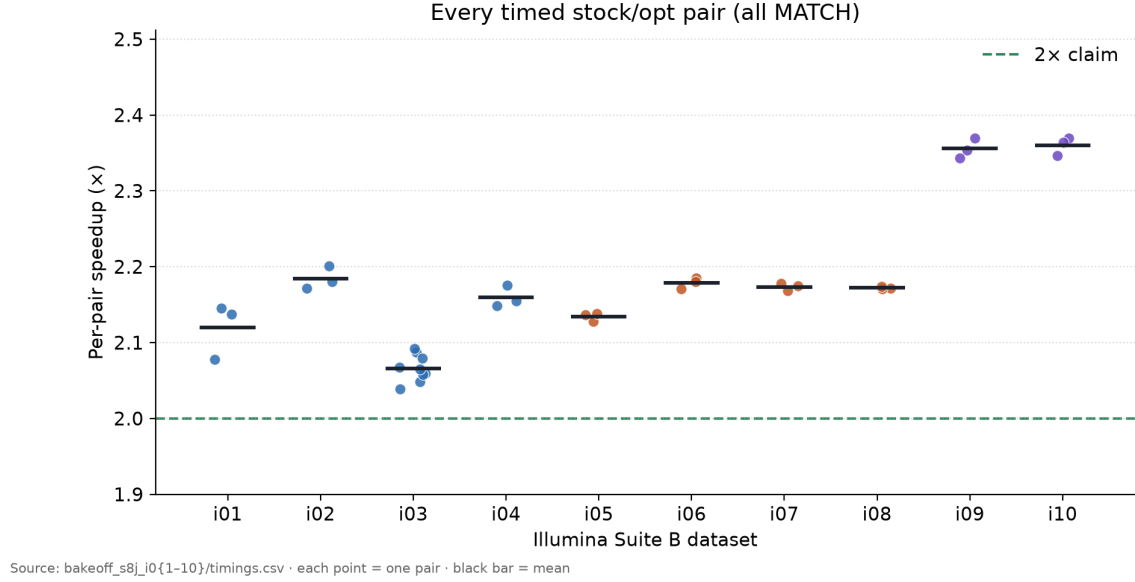


Figure 5: Every timed stock/ACTS pair on Suite B. Black bars are per-dataset means. All pairs are MATCH and above 2 $\times$.

5.3 Threads, machines, and build extras

All graded timings use one mapping thread. Multi-threaded runs share the same ACTS patch but were not the acceptance criterion; contention and BAM sorting can move the ratio. Absolute seconds on Apple Silicon will not match an x86 node. The fairness claim is stock versus ACTS on one machine, with the same flags. Profile-guided optimization, link-time optimization, and jemalloc are part of the shipped binary. They do not alter Definition 1. A reader who wants a pure algorithmic ablation can rebuild without them; the rung ladder in Figure 2 already separates algorithmic stacking from the final jemalloc+PGO step on i01.

5.4 Withdrawn baseline

The oversized-index configuration (`genomeSAindexNbases = 14` on a 10 Mb reference, with an 800-pair read slice) reported an apparent $\sim 2.5\times$ speedup that was mostly avoided `SAindex` load. Figure 9 keeps that number visible as a negative control. It is not part of the Suite B claim. Honest indexes follow STAR’s recommended Nbases; honest read sets are the Suite B FASTQs in Section 4.2.

5.5 Relation to other aligners

HISAT, Bowtie 2, and BWA solve related but different problems or use different indexes [5, 6, 8]. A bake-off against those tools under free flag choice would answer a different question: which aligner is fastest for a lab’s tolerance for alignment differences. This paper answers only whether a drop-in STAR binary can be observationally equivalent and at least twice as fast on the locked Suite B protocol.

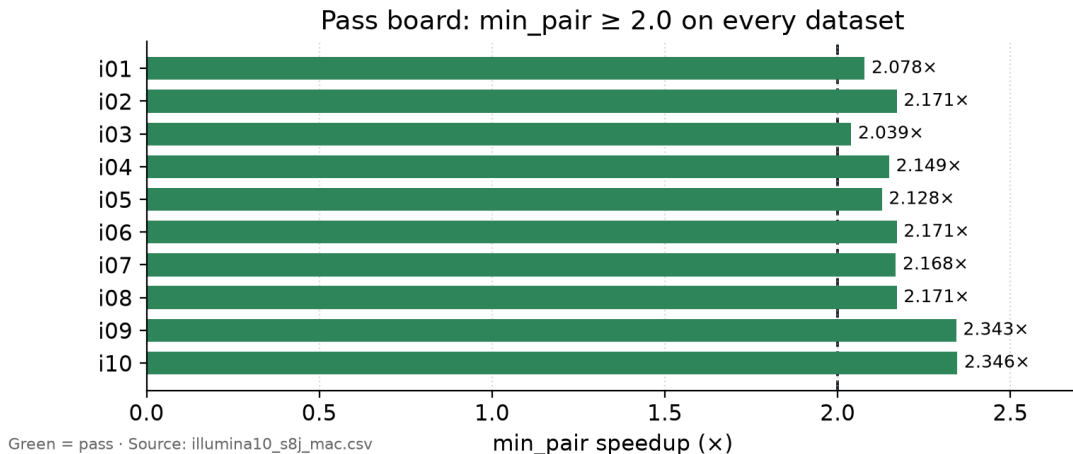


Figure 6: Pass board: $\rho^{\min}$ for each dataset. Green bars are sets with $\rho^{\min} \geq 2$.

6 Conclusion

ACTS accelerates spliced alignment in STAR by amortizing candidate transcript state inside the stitch recursion, without changing the alignments that exit the tool. We defined observational equivalence for drop-in binaries, cast stock’s per-node transition as a full-state copy followed by Stitch, and replaced that copy with bounded exon transfers, in-place mutation with constant-time undo, dead-stitch skipping, leaf adoption, and closed-form match scoring. Under a locked single-thread command line on ten Illumina Suite B sets, every timed pair matched stock SAM after header normalization, and every dataset’s worst pair was at least 2.039 $\times$ faster (i03). Mean per-pair speedups ranged from 2.066 $\times$ to 2.360 $\times$.

The result is scoped. It holds on stitch-bound teaching indexes and does not license a free comparison against other aligners or against multi-threaded full-genome production runs. An oversized-index configuration that appeared to deliver $\sim 2.5\times$ is withdrawn; honest Nbases and MATCH-gated timings are part of the claim. Build extras (PGO, LTO, jemalloc) ship with the binary but are not a new alignment algorithm.

The practical takeaway is narrow and actionable: when a production aligner’s hot path spends first-order time copying candidate state that the search does not need, observational equivalence is a useful contract for removing that cost. ACTS is one instance of that contract on STAR 2.7.11b. Code, Suite B fetch scripts, and figures are available at <https://github.com/thatrandomasiandev/csci270-research-project>.

Acknowledgments

We thank Professor Jiapeng Zhang, PhD, for locking the evaluation criteria that forced an honest index, identical flags, and byte-level output comparison. Upstream STAR is due to Dobin and colleagues [2, 3].

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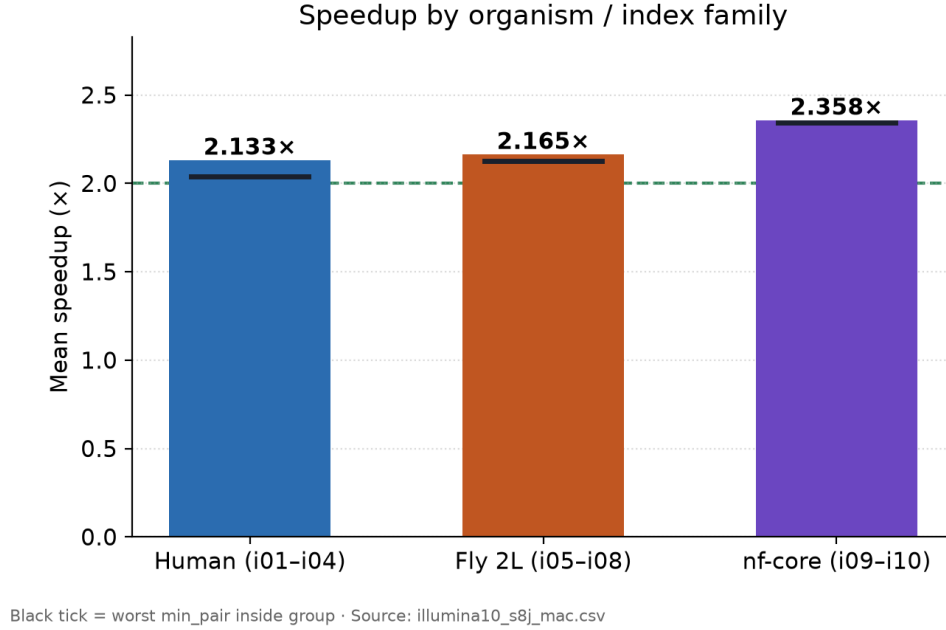


Figure 7: Speedup by organism and index family. Bars are group means of ρ^{mean} ; ticks are the worst ρ^{min} inside the group.

scheduling and inlining but do not remove redundant large-object copies; cited as background for build extras.

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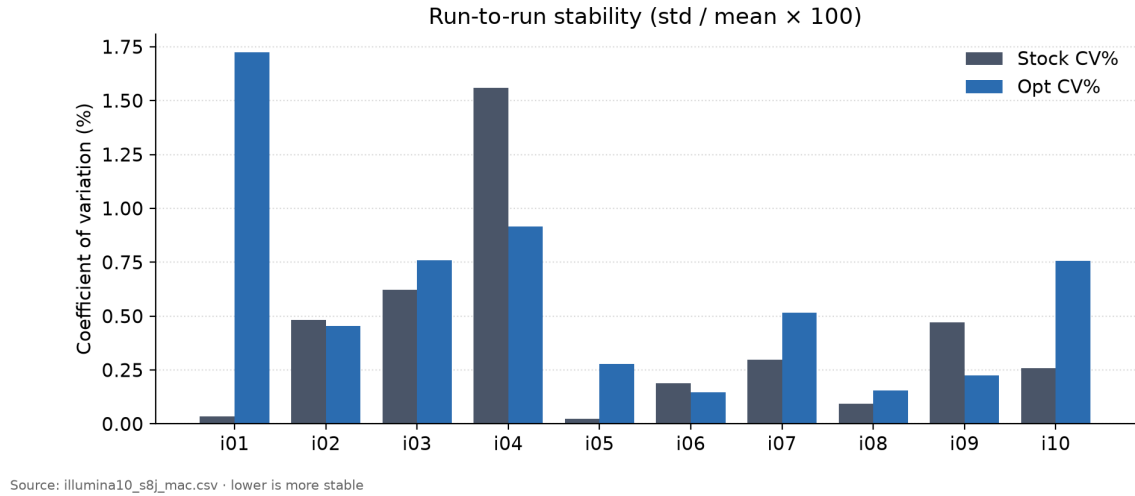


Figure 8: Coefficient of variation (%) for stock and ACTS wall-clock across timed runs.

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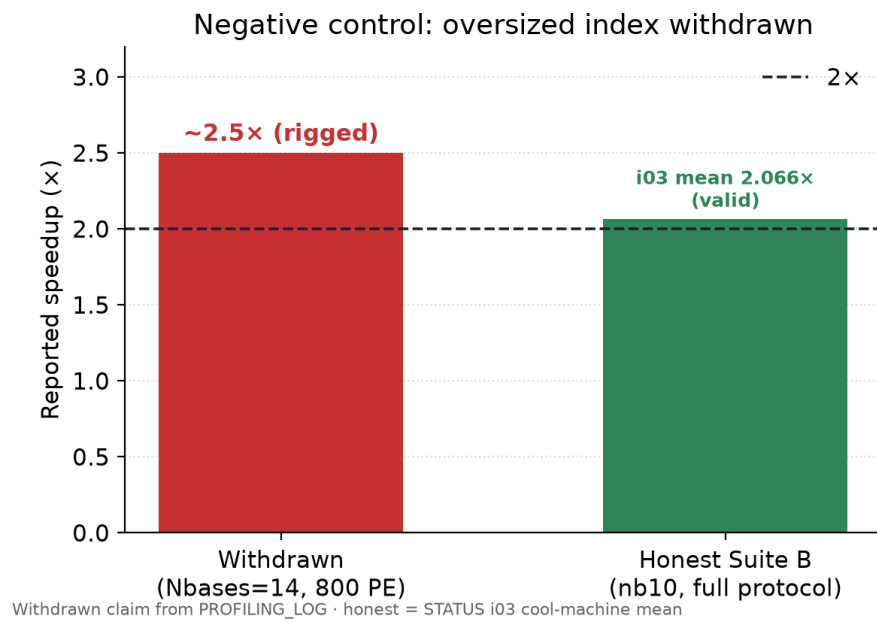


Figure 9: Negative control: withdrawn oversized-index headline versus the honest i03 mean under recommended Nbases.